

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Architected AWS serverless infrastructure (Lambda, S3, DynamoDB, API Gateway, SageMaker) exposing production APIs at sub-200ms latency, supporting high-throughput distribution at scale.
- Owned a deployment-gated evaluation pipeline enforcing automated quality, latency, and drift checks before every release, blocking 15% of releases that failed validation criteria in production.
- Led design of a multi-agent system using OpenAI Agents SDK, Claude, and MCP tool-calling, applying AI-native development tools daily to accelerate delivery on production workflows handling millions of daily transactions.

QureAI

Jan 2024 – May 2025

ML Infrastructure Engineer

- Drove a RAG retrieval redesign improving recall 61% to 89% on a 2K-query eval set via hybrid dense-sparse retrieval and re-ranking.
- Led migration of synchronous model serving to async vLLM on GCP Cloud Run with autoscaling, cutting p99 inference latency 420ms to 140ms across 50K+ daily requests.

Nomura Research Institute Financial Technologies

Aug 2020 – Aug 2023

Software Engineer

- Architected and operated 25+ production Java microservices exposing REST APIs for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS) in a distributed, fault-tolerant architecture.
- Drove a redesign of nightly financial data aggregation on SQL/PostgreSQL, cutting downstream-blocking processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Led standardization of failure-handling contracts across 25+ services, incorporating perspectives from cross-functional teams across time zones, reducing production escalations 60% in one quarter through code reviews and design write-ups.

Projects

Distributed KV Store / VPC Control Plane Simulator | github.com/ganeshhgupta/distributed-kv-store

- Built a Raft-consensus key-value store with leader election, log replication, and log compaction; companion VPC control plane simulator modeling subnet routing and security group enforcement for fault-tolerant distributed coordination.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production service with rubric-based scoring, drift detection, and per-call latency tracking across model versions; FastAPI backend, PostgreSQL.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Java, Python, SQL, TypeScript

Backend & APIs: REST APIs, Web Services, Microservices, PostgreSQL, FastAPI, Node.js

Distributed Systems: Fault Tolerance, High Availability, Consensus Protocols (Raft), Capacity Management

Cloud & Ops: AWS (Lambda, EC2, S3, RDS, SQS, SageMaker), GCP (Cloud Run), Docker, Kubernetes, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer